

Module -5

Fundamental Tie-set Matrix

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SENSE

Tie-Set Matrix (Loop Matrix)

- A tie – set is a set of branches contained in a loop such that each loop contains one link and the remainder are tree branches.

(or)

- The set of branches forming the closed loop in which link or loop circulate is called a Tie-set.
- From the knowledge of tie-sets (basic loops), the loop matrix can be formed.
- In this matrix, the loop orientation is to be the same as corresponding link direction.

Steps to draw Tie-set Matrix

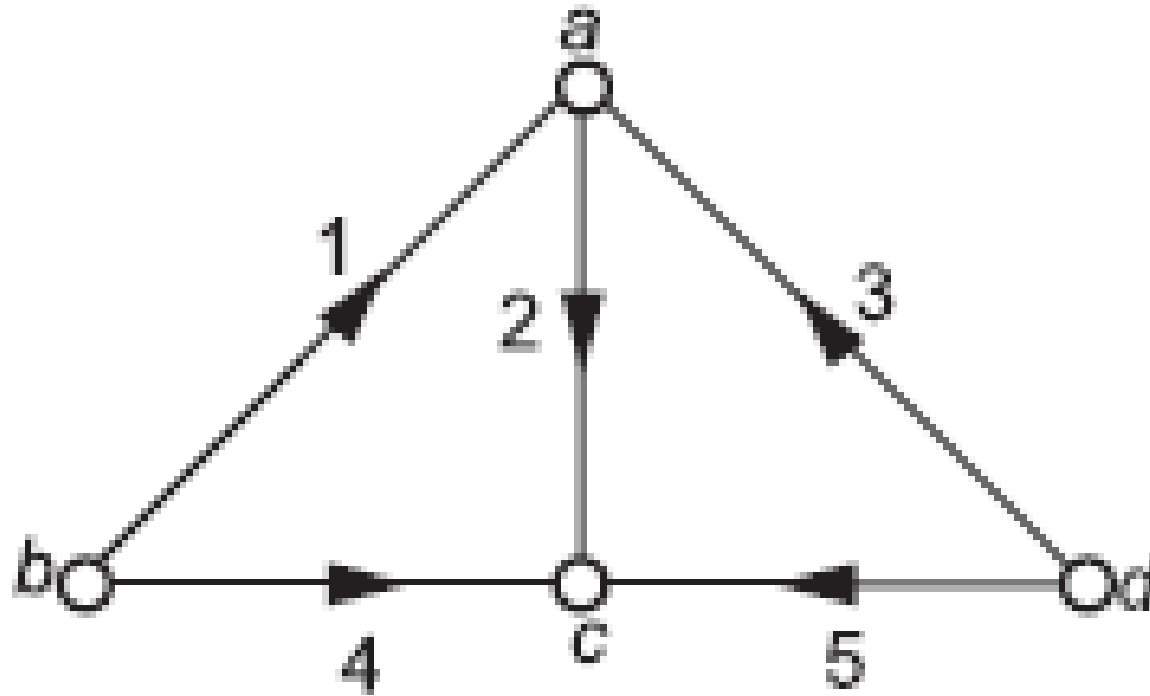
1. Draw the oriented graph of the network. Choose a tree.
2. Each link forms an independent loop. The direction of this loop is same as that of the corresponding link. Choose each link in turn.
3. Prepare the tie-set matrix with elements $b_{i,j}$,

where $b_{i,j} = 1$, when branch b_j in loop i and is directed in the same direction as the loop current.
 $= -1$, when branch b_j is in loop i and is directed in the opposite direction as the loop current.
 $= 0$, when branch b_j is not in loop i .

Tie-set matrix is an $i \times b$ matrix.

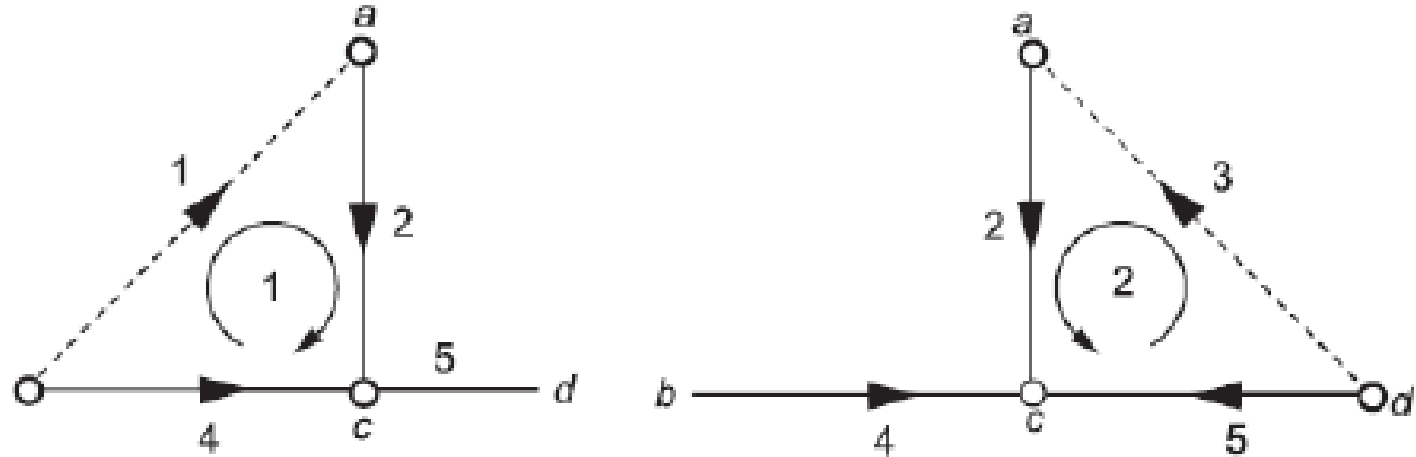
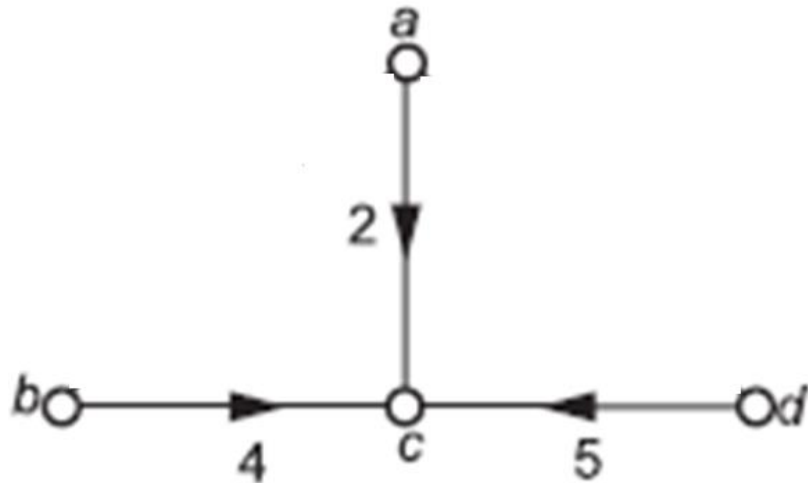
Numerical - 1

- Form the tie-set matrix for the graph shown.



Solution:

Tree



Tie-set Schedule

No. of Branch currents (j)

No. of links (i)	1	2	3	4	5
1	1	1	0	-1	0
2	0	1	1	0	-1

$$\mathbf{B} = \begin{bmatrix} 1 & 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & -1 \end{bmatrix}$$

The columns of B matrix gives the branch currents in terms of link currents.

Branch currents are represented as “j” and link currents are represented as “i”.

$$J_1 = i_1$$

$$J_2 = i_1 + i_2$$

$$J_3 = i_2$$

$$J_4 = -i_1$$

$$J_5 = -i_2$$

$$\left. \begin{array}{l} J_1 = i_1 \\ J_2 = i_1 + i_2 \\ J_3 = i_2 \\ J_4 = -i_1 \\ J_5 = -i_2 \end{array} \right\} \text{Eqn. 1}$$

The rows of the schedule give KVL equations in terms of the coefficient of the schedule.

$$V_1 + V_2 - V_4 = 0$$

$$V_2 + V_3 - V_5 = 0$$

$$\left. \begin{array}{l} V_1 + V_2 - V_4 = 0 \\ V_2 + V_3 - V_5 = 0 \end{array} \right\} \text{Eqn. 2}$$

- $J_B = [B^T][I_L]$

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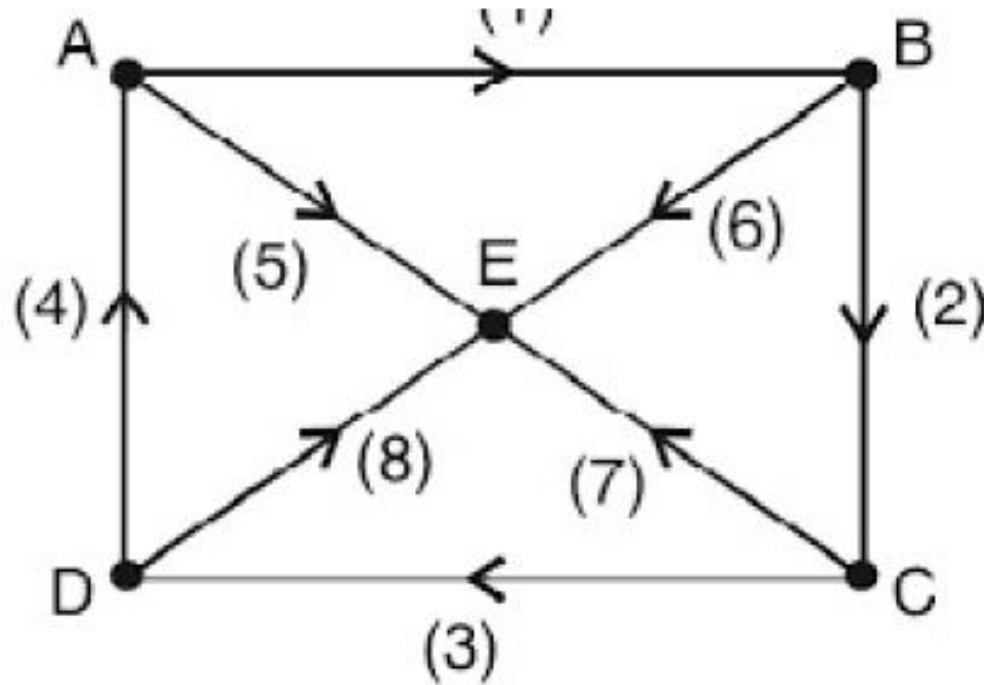
$$\begin{pmatrix} J1 \\ J2 \\ J3 \\ J4 \\ J5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} I1 \\ I2 \end{pmatrix}$$

- $[B] [V_B] = 0$

$$\begin{bmatrix} 1 & 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & -1 \end{bmatrix} \begin{pmatrix} V1 \\ V2 \\ V3 \\ V4 \\ V5 \end{pmatrix} = 0$$

Numerical - 2

- Form the tie-set matrix for the graph shown.



Numerical - 3

Form the tie-set matrix for the graph shown.

